COLOSSUS 2.0

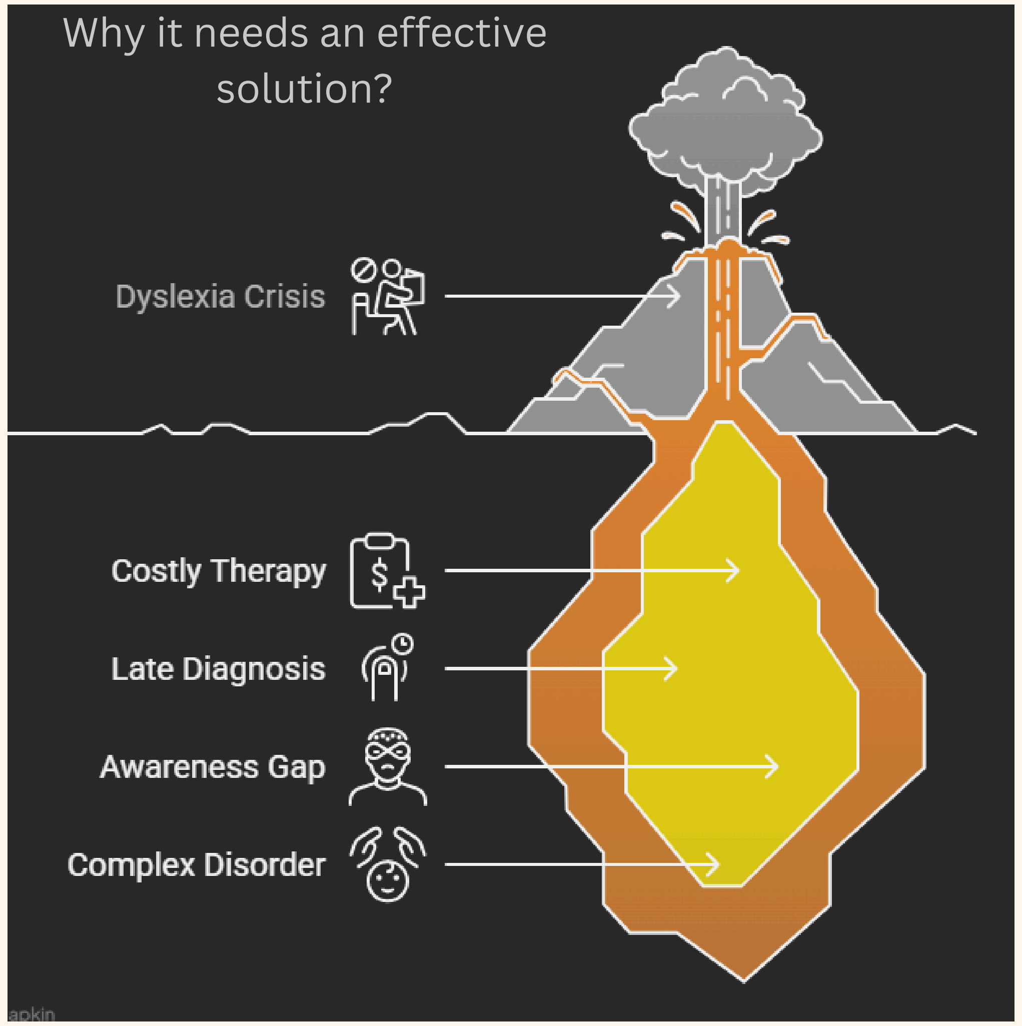
TEAM DETAILS

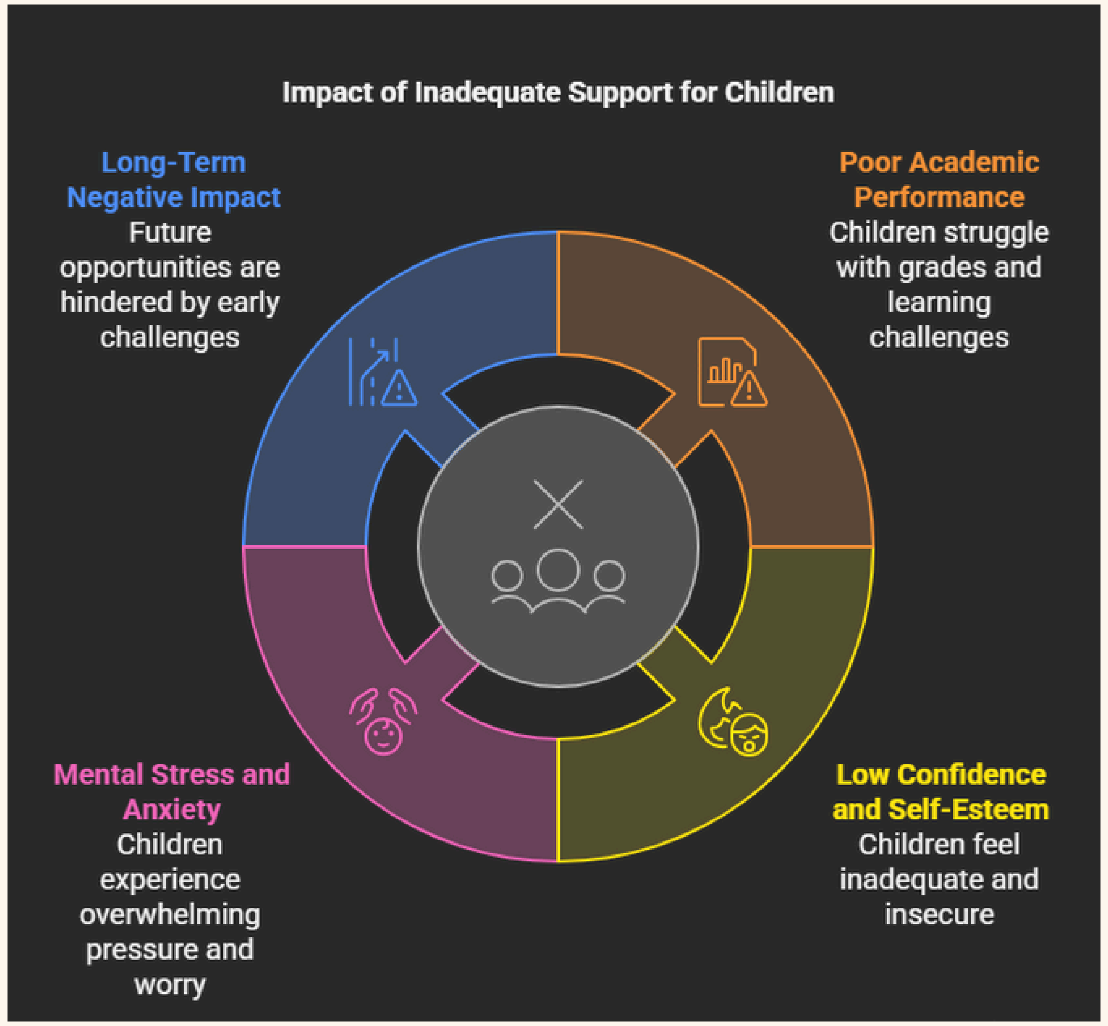
MindBridge : Path to Cognitive Confidence

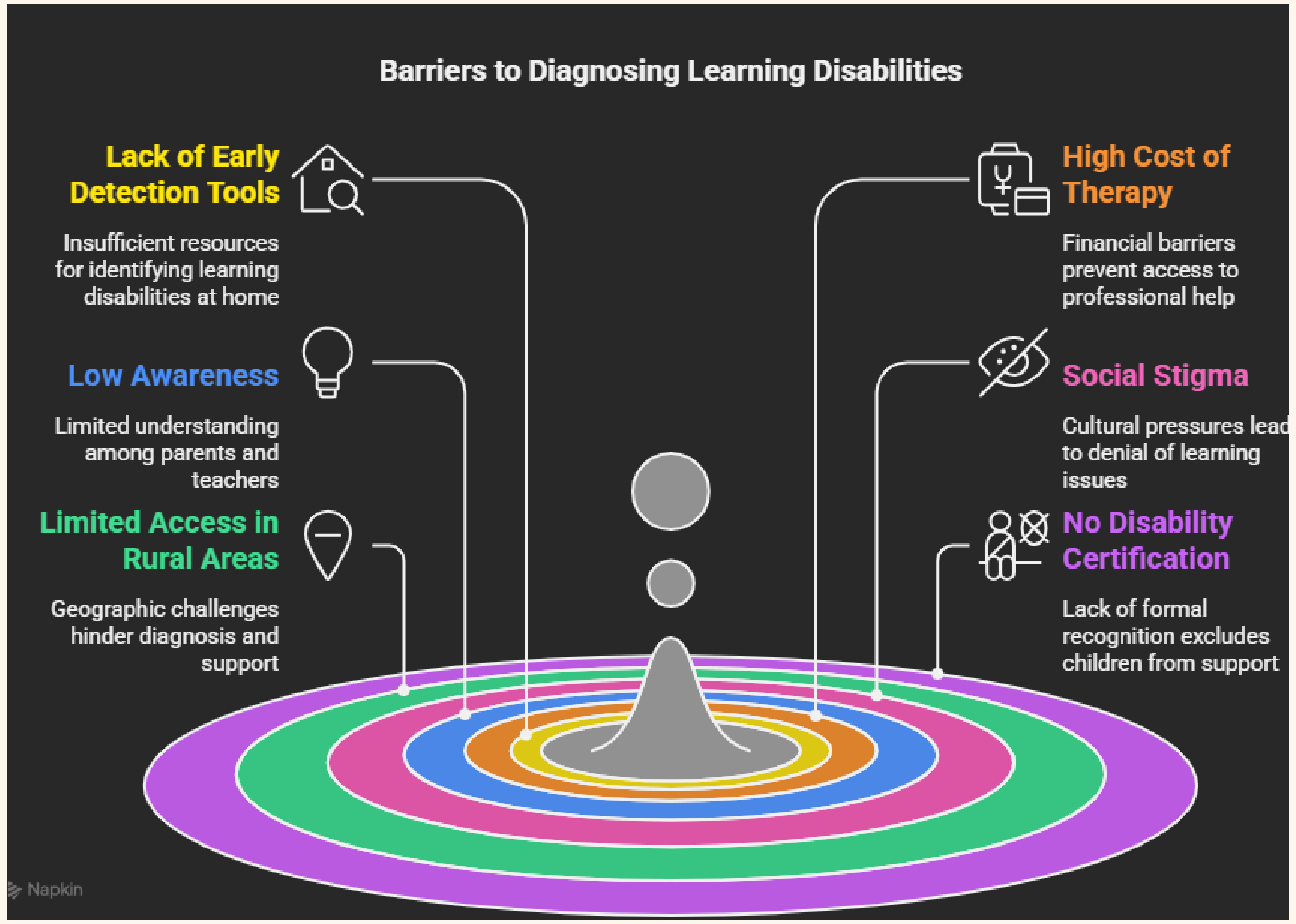
- Team Name: CocoCoders
- Team Leader Name: Rohan Jaiswal
- Track: Education and Healthcare

PROBLEM STATEMENT

- Specific Learning Disabilities (SLD) like Dyslexia, Dyscalculia, and Dysgraphia are neurological conditions, not intelligence problems.
- Without timely support, affected children struggle in reading, writing, or math — leading to low grades, low confidence, and emotional distress.
- These children are often misunderstood, punished, or labeled as "lazy" instead of being helped.
- Globally, 10–15% of school-age children have some form of SLD. In India alone, over 35 million children are estimated to be affected.
- Most go undiagnosed, especially in middle-class and rural families, due to stigma, cost, and lack of access to specialists.

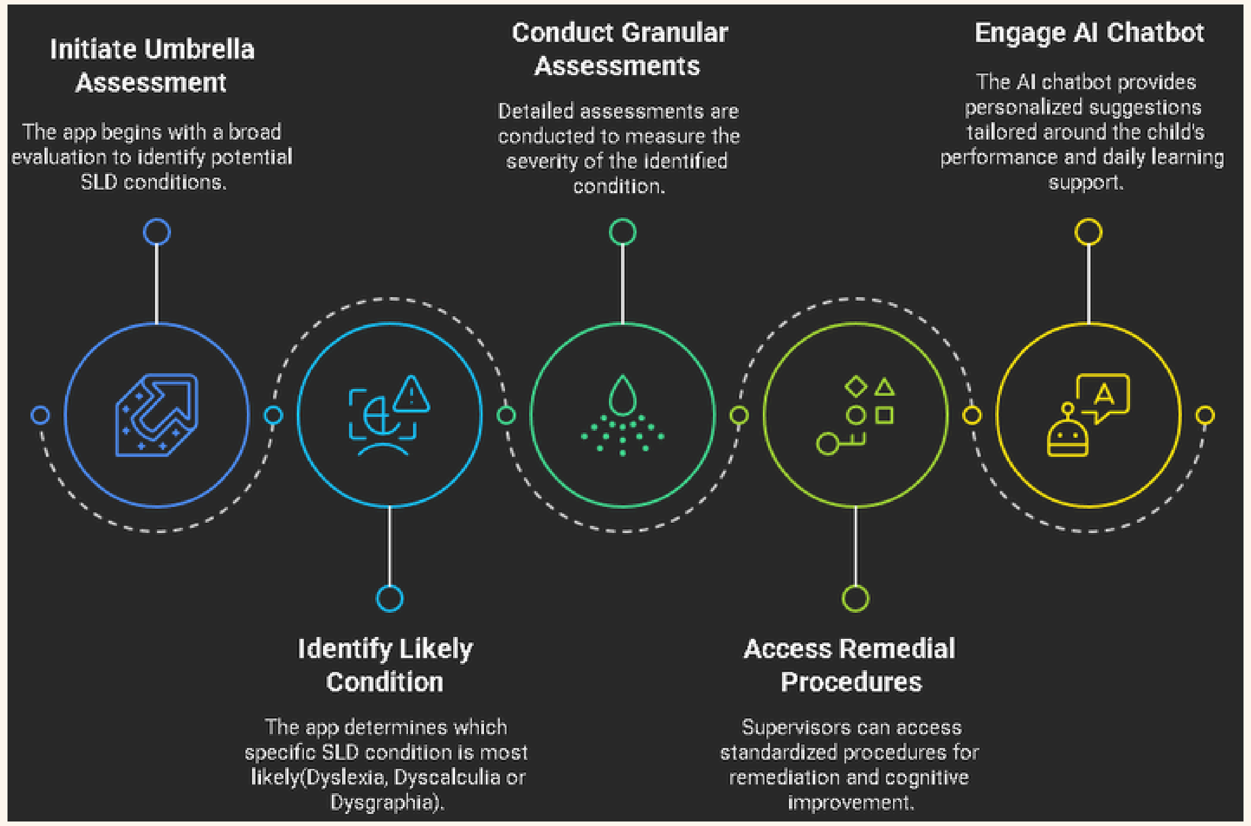






IDEA

- A mobile app designed to detect and support children with Specific Learning Disabilities (SLD) — Dyslexia, Dysgraphia, and Dyscalculia.
- Targets early detection by allowing parents/supervisors to screen children at home.
- Starts with an umbrella assessment to identify the most affected area.
- Follows with detailed, condition-specific assessments to measure severity.
- Provides standardized remedial strategies that parents can follow for home-based support.
- Includes an AI-powered chatbot that acts like a virtual special educator, personalized to the child's progress.
- Offers daily tips, learning plans, and actionable guidance for parents based on child's needs.
- Aims to make screening and treatment affordable, accessible, and effective — especially for middle-class and rural families.



Early detection is rare due to lack of awareness, high diagnosis costs, and limited access to specialists—especially in rural and middle-class communities.

Families cannot afford professional assessments and therapy sessions, which are often expensive and ongoing.

Parents often dismiss the signs or are in denial, believing their child will "catch up" with time.

Without proper certification or diagnosis, students are denied accommodations in schools and exams.

Children are left undiagnosed, untreated, and misunderstood—leading to poor academic performance, low self-esteem, and social isolation.

There is a lack of personalized, at-home guidance for parents who want to help their child improve.

KEY FEATURES

🔍 **Smart SLD Screening (Umbrella Classification)**
- A quick, AI-based questionnaire helps identify which type of learning difficulty (Dyslexia, Dyscalculia, Dysgraphia) the child is likely experiencing.
- Output: A visual chart showing percentage likelihood across all three areas to guide further action.

🧠 **Granular Assessments**
- Based on umbrella results, the app guides the user through deeper, specific assessments in the identified category:
  - ✅ Dyslexia: Phonological awareness, rhyming, decoding, reading fluency & comprehension.
  - ✍️ Dysgraphia: Assess handwriting legibility, formation.
  - 🔢 Dyscalculia: Evaluate number sense, arithmetic skills, and pattern recognition.
- All responses are analyzed via audio, image (OCR), and logic-based inputs to give accurate results.

📊 **AI-Generated Severity Scoring**
- For each category, the system calculates a severity level based on detailed metrics like WER (Word Error Rate), response time, confidence score, etc.
- Helps parents/supervisors understand the depth of difficulty.

🎯 **Personalized Recommendations**
- Based on assessment scores, the app provides:
  - Daily learning tips
  - At-home exercises
  - Remedial action plans aligned with severity

🤖 **Interactive Support Chatbot (Works Like a Special Educator)**
- A centralized AI chatbot supports all three subcategories.
- Provides guidance, learning strategies, doubt-solving, and reminders personalized to the child's performance.
- Makes the app feel like a 24/7 companion educator for the parent or supervisor.

🔄 **Feedback-Based Adaptive Learning**
- The app continuously learns from user performance and feedback to refine future assessments and recommendations.
- Personalized suggestions evolve over time based on the child's progress and input from the parent/supervisor.

📚 **Motivational Blogs & Awareness Content**
- The app features a dedicated section for blogs, stories, and articles focused on learning disabilities.
- These include motivational success stories, expert insights, and awareness pieces that help parents and supervisors better understand SLD.
- The goal is to reduce stigma and promote the idea that learning differences do not define a child's intelligence or potential.

📍 **Nearby Support Resources**
- The app provides a curated list of nearby special education centers, therapists, and inclusive schools based on the user's location.
- This helps parents/supervisors easily access professional help if needed, bridging the gap between digital and real-world support.

📄 **Certification & Reports**
- Downloadable, well-structured reports post-assessment — which can be shown to professionals or schools if needed.

🎓 **Parent Learning Hub**
- Micro-courses or tutorials for supervisors to understand SLD, effective teaching strategies, communication tips, and do's & don'ts for supporting their child.

TECH STACK

Flutter

Purpose

- Cross-platform mobile development framework.
- Allows you to build native-like apps for Android and iOS from a single codebase.

Benefits

- Fast development with Hot Reload.
- Rich widget library for smooth, responsive UIs.

Firebase

Purpose

- Provides backend services like user authentication, database, cloud storage, and serverless functions.

Benefits

- Fully managed, scalable backend — ideal for startups and prototypes.
- Real-time updates using Firebase Realtime Database or Firestore (NoSQL).

Whisper

Purpose

- Converts audio inputs (child's speech) into text for assessment analysis.

Benefits

- High accuracy in noisy environments..
- Whisper (open-source) allows offline and on-device processing.

Google Places API

Purpose

- Used to show nearby special education centers and therapists.

Benefits

- Accurate, location-aware recommendations.
- Integrates seamlessly with Flutter using google_maps_flutter plugin.

OpenCV PyTorch

Purpose

- Analyze handwriting for legibility, pressure, spacing, line alignment.

Benefits

- OpenCV extracts contours, alignment, and structure.
- Neural networks score handwriting characteristics against a trained model.

Python

Purpose

- Analyzes raw input (text, audio, image) and computes severity using metrics like WER, response time, transcription confidence, etc.

Benefits

- Highly customizable and integrates with ML-based feedback loops.

Express JS docker

Purpose

- Backend Reporting Engine: Express.js + Puppeteer with Docker customization : Dynamically generate downloadable PDF reports of assessment results and progress.

Benefits

- Highly customizable and integrates with ML-based feedback loops.

Gemini

Purpose

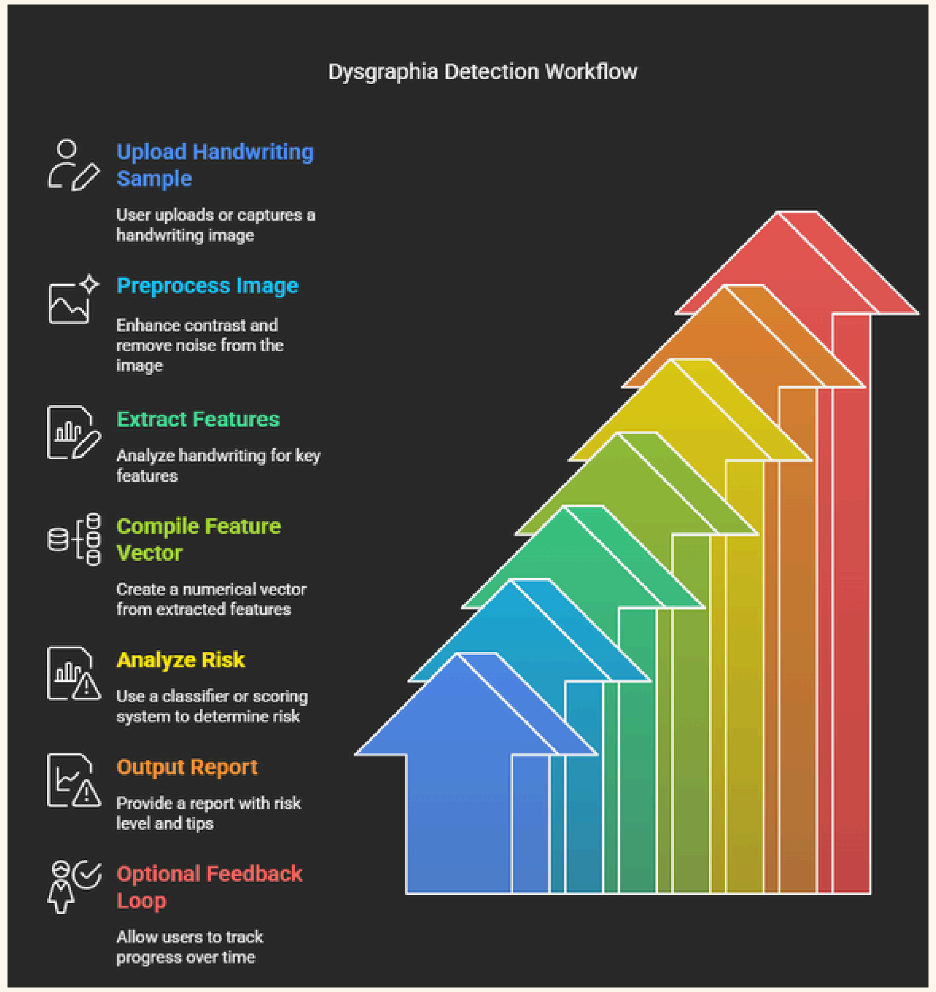
- To act as a 24/7 virtual special educator and personal assistant for parents/supervisors throughout the child's SLD journey.

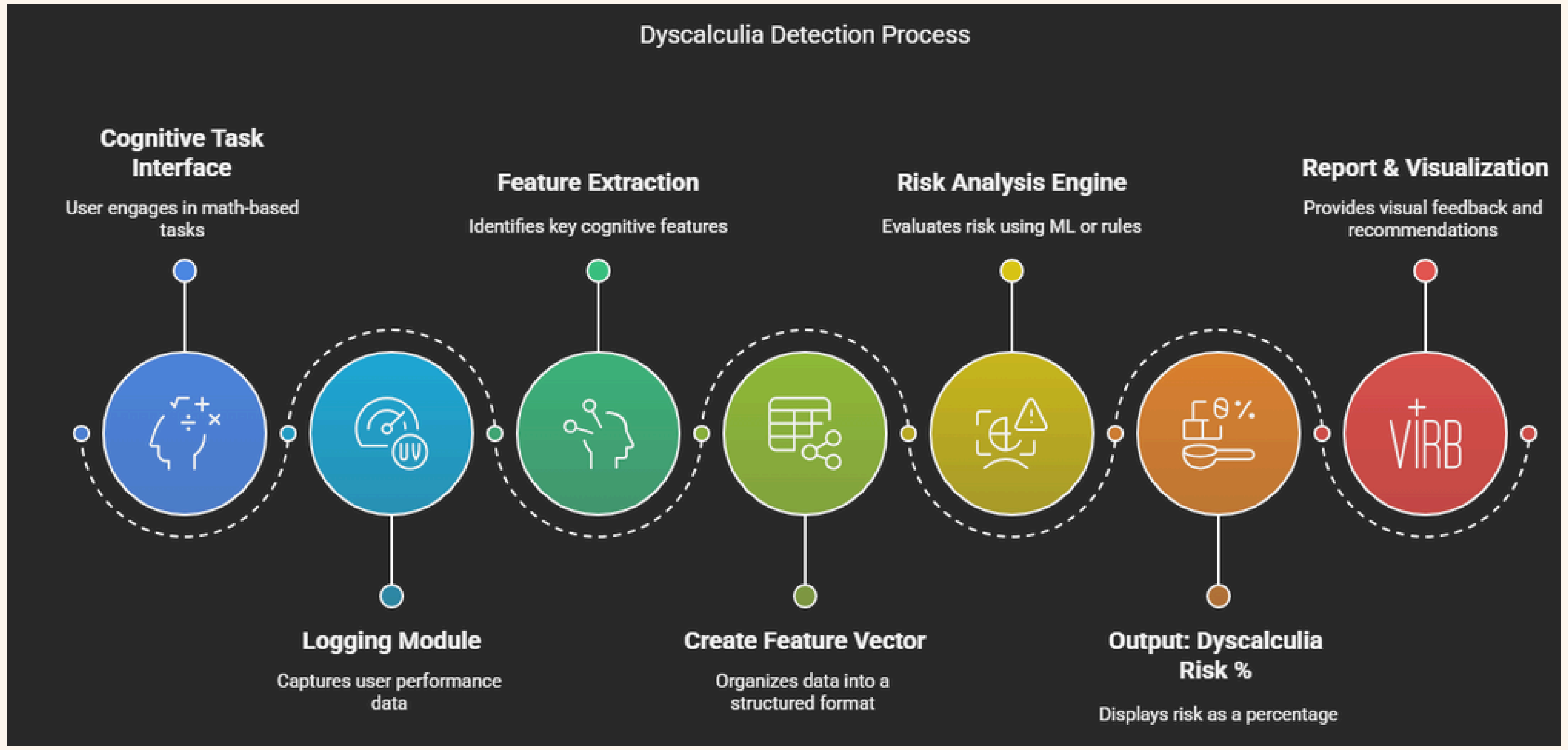
Benefits

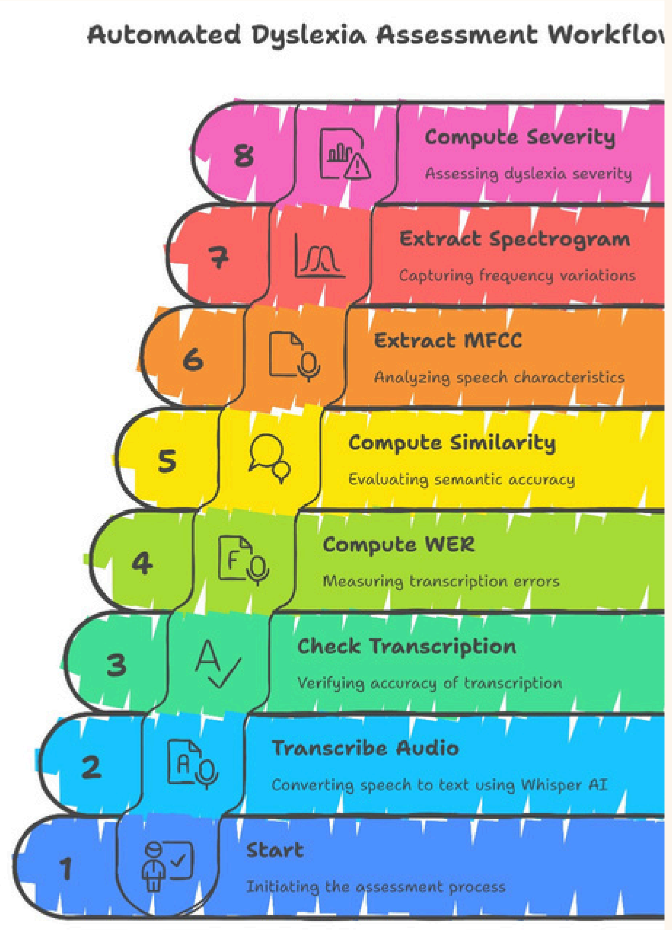
- Offers guidance and learning strategies for dyslexia, dysgraphia, and dyscalculia — all from a single, cohesive assistant.
- Eliminates the dependency on human educators or therapists for daily support. Parents can get help anytime, anywhere tailored to the child's performance and helps in decision making.

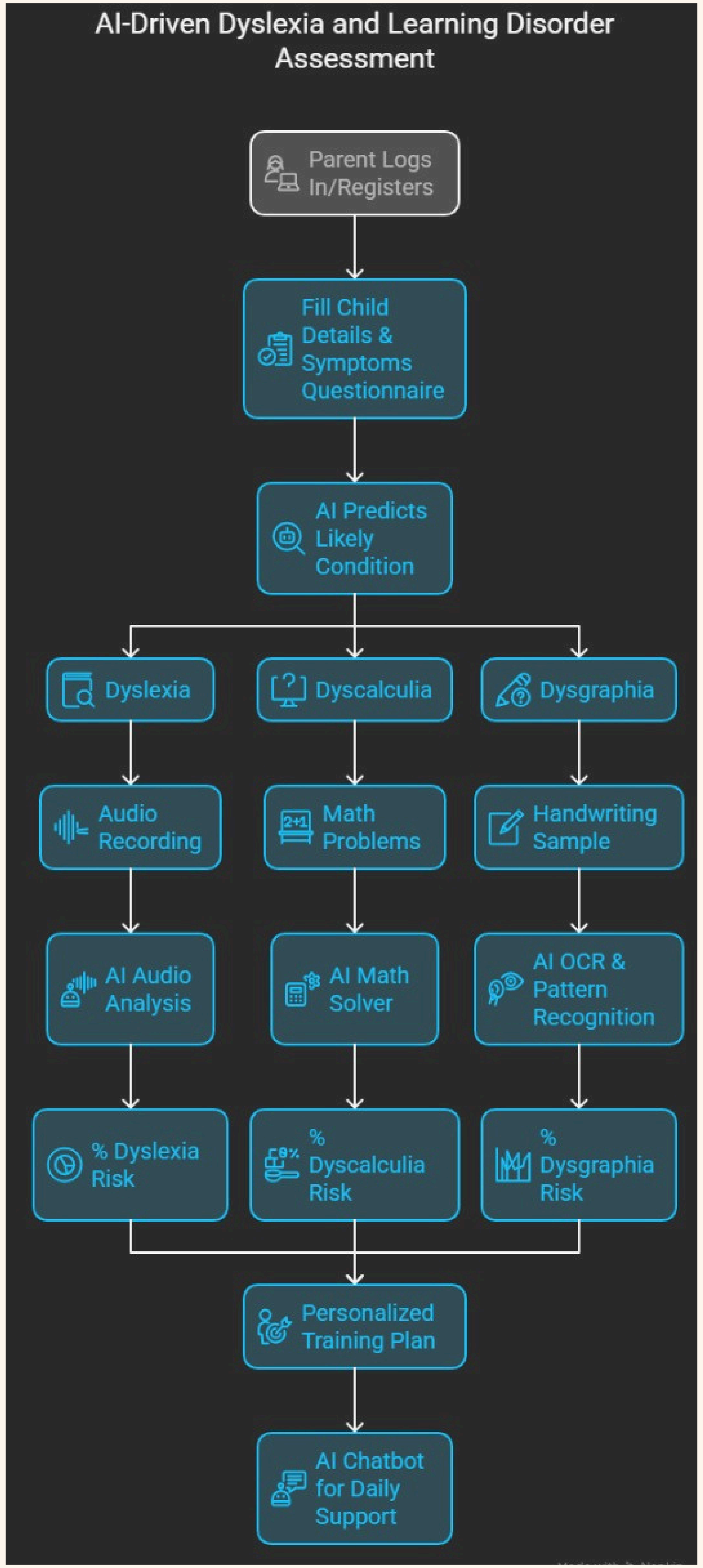
FLOW DIAGRAM

1. 🔐 **User Authentication & Role Selection**
   The app starts with a login screen where users (parents or guardians) sign in and select the child's profile for assessment.
2. 📋 **Initial Questionnaire Module**
   A set of screening questions is provided to gather background information on the child's learning habits, school performance, and daily behavior.
3. 🧠 **Multi-Module Diagnostic Input Phase**
   Based on the suspected condition, the app activates one or more of the following input modules:
   - 🎤 **Dyslexia** → Child reads a passage aloud (audio is recorded).
   - ✍️ **Dysgraphia** → Upload or capture an image of a handwriting sample.
   - 🔢 **Dyscalculia** → Child solves math problems and memory tasks in-app.
4. ⚙️ **Preprocessing and Feature Extraction**
   - Audio is cleaned and transcribed (Dyslexia).
   - Images are enhanced and processed (Dysgraphia).
   - Math response timing and accuracy are logged (Dyscalculia).
   - Features like WER (Word Error Rate), spelling/letter errors, response time, pressure (if on tablet), and memory scores are extracted.
5. 📊 **AI-Based Analysis Engine**
   Each module passes its features to the AI engine:
   - Uses ML classifiers (e.g., Random Forest, XGBoost, SVM) or rule-based logic.
   - Outputs the **risk percentage** for each learning disability.
   - Categorizes severity into **Low, Moderate, or High**.
6. 📄 **Report Generation & Visualization**
   A visual report is generated showing:
   - Percentage risk of each condition
   - Highlighted errors or difficulties
   - Suggested exercises or next steps
7. 📚 **Learning Recommendations & Resources**
   Based on the child's results, the app provides:
   - Exercises or games for practice
   - Tips for parents
   - Options to connect with special educators or therapists
8. 🔄 **Feedback Loop**
   Users can re-test over time and monitor progress. The app logs performance history for comparison and improvement tracking.
9. 💬 **Chatbot Assistant**
   A friendly AI chatbot helps explain the results, guide users through next steps, and answer common questions like a special educator would.

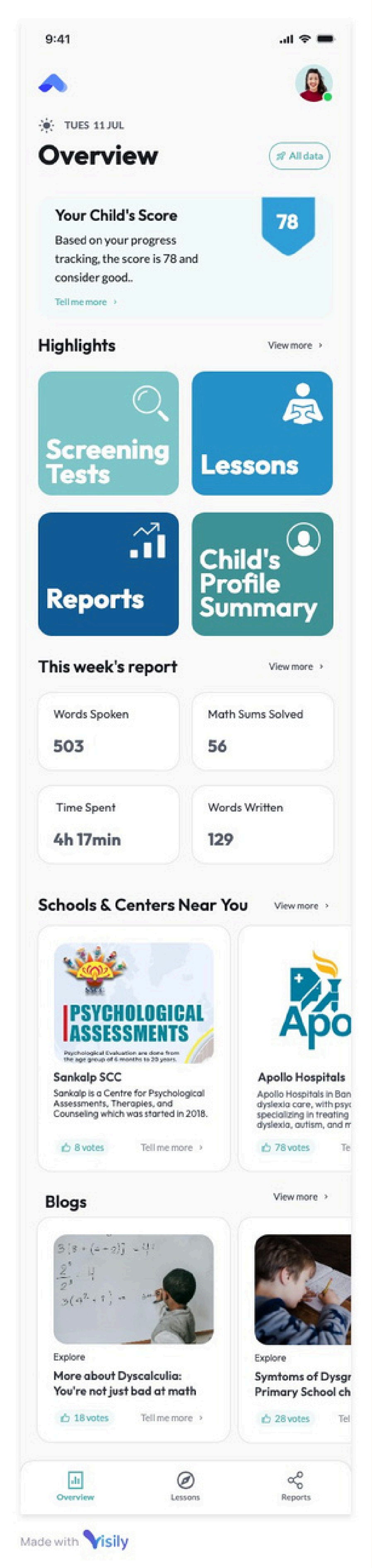


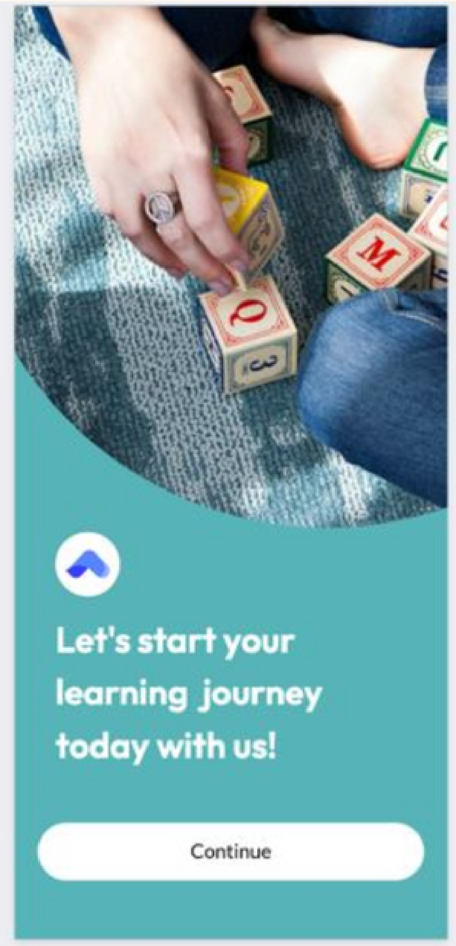


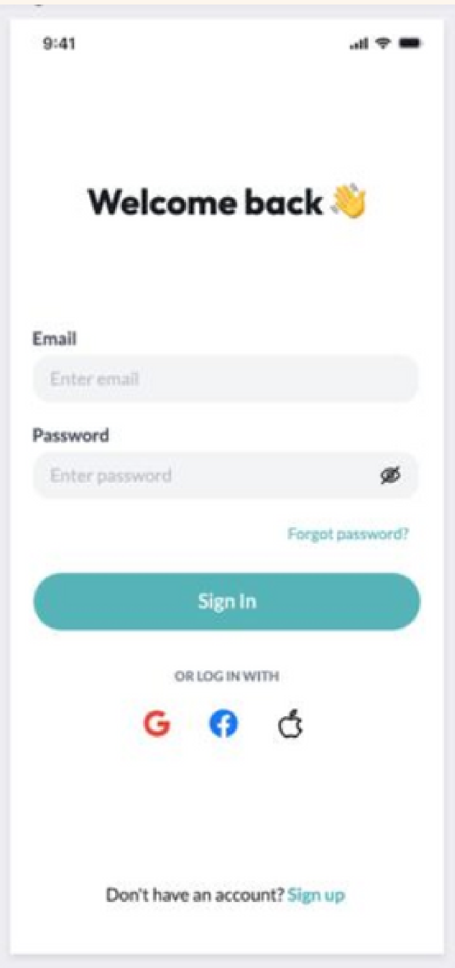


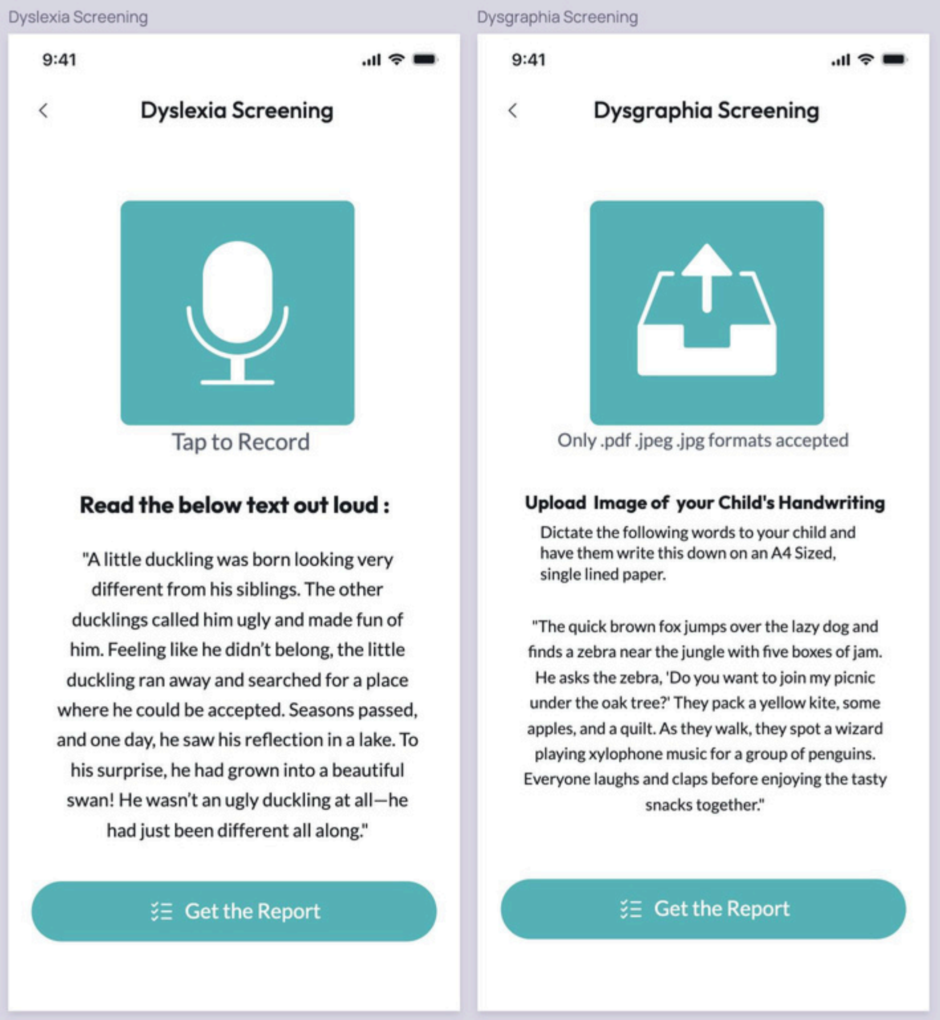


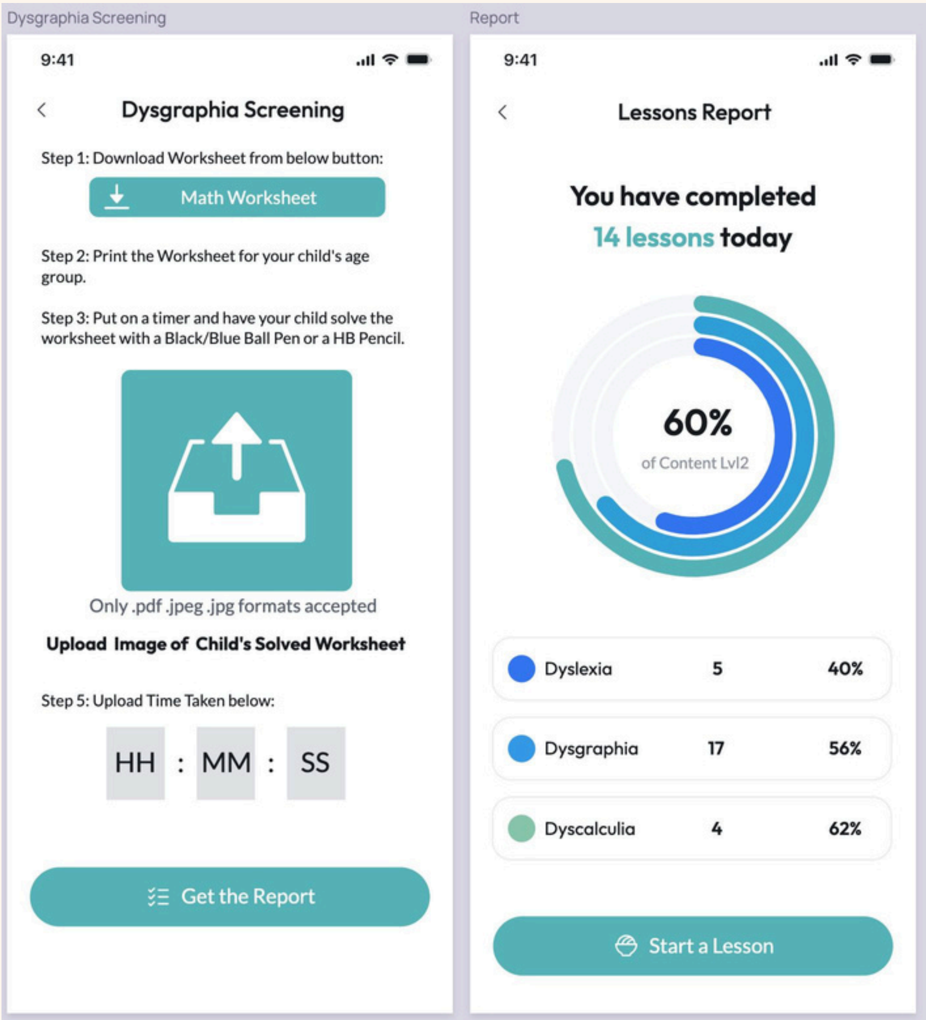
PROTOTYPE

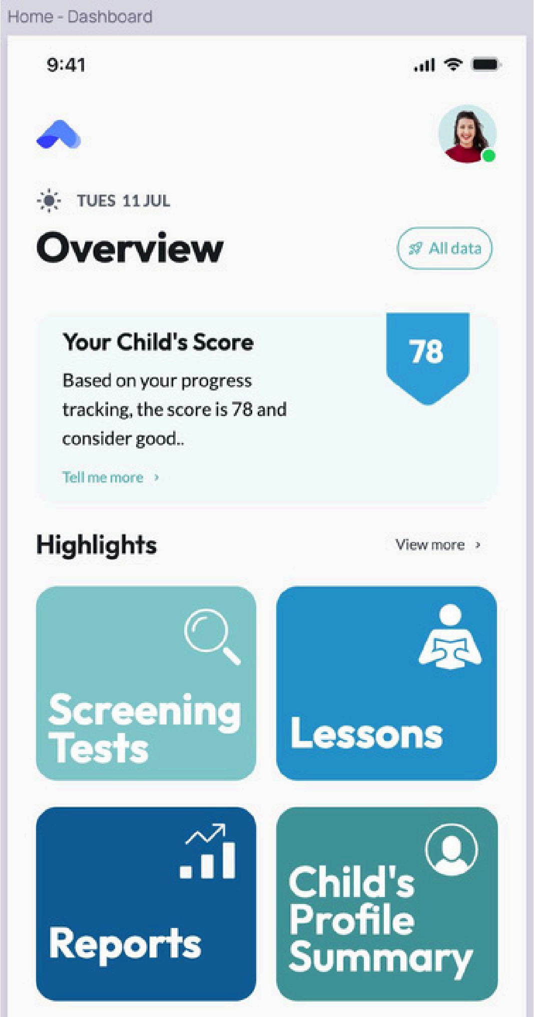

9:41
TUES 11 JUL
Overview
All data
Your Child's Score
78
Based on your progress tracking, the score is 78 and consider good..
Tell me more
Highlights
View more
Screening Tests
Lessons
Reports
Child's Profile Summary
This week's report
View more
Words Spoken
503
Math Sums Solved
56
Time Spent
4h 17min
Words Written
129
Schools & Centers Near You
View more
PSYCHOLOGICAL ASSESSMENTS
Sankalp SCC
Sankalp is a Centre for Psychological Assessments, Therapies, and Counseling which was started in 2018.
8 votes
Tell me more
Apollo Hospitals
78 votes
Blogs
View more
Explore
More about Dyscalculia: You're not just bad at math
18 votes
Tell me more
Explore
28 votes
Overview
Lessons
Reports
Made with Visily


Let's start your learning journey today with us!
Continue


9:41
Welcome back 👋
Email
Enter email
Password
Enter password
Forgot password?
Sign In
OR LOG IN WITH
Don't have an account? Sign up


Dyslexia Screening
Dysgraphia Screening
9:41
Dyslexia Screening
Tap to Record
Read the below text out loud :
"A little duckling was born looking very different from his siblings. The other ducklings called him ugly and made fun of him. Feeling like he didn't belong, the little duckling ran away and searched for a place where he could be accepted. Seasons passed, and one day, he saw his reflection in a lake. To his surprise, he had grown into a beautiful swan! He wasn't an ugly duckling at all—he had just been different all along."
Get the Report
9:41
Dysgraphia Screening
Only .pdf .jpeg .jpg formats accepted
Upload Image of your Child's Handwriting
Dictate the following words to your child and have them write this down on an A4 Sized, single lined paper.
"The quick brown fox jumps over the lazy dog and finds a zebra near the jungle with five boxes of jam. He asks the zebra, 'Do you want to join my picnic under the oak tree?' They pack a yellow kite, some apples, and a quilt. As they walk, they spot a wizard playing xylophone music for a group of penguins. Everyone laughs and claps before enjoying the tasty snacks together."
Get the Report


Dysgraphia Screening
Report
9:41
Dysgraphia Screening
Step 1: Download Worksheet from below button:
Math Worksheet
Step 2: Print the Worksheet for your child's age group.
Step 3: Put on a timer and have your child solve the worksheet with a Black/Blue Ball Pen or a HB Pencil.
Only .pdf .jpeg .jpg formats accepted
Upload Image of Child's Solved Worksheet
Step 5: Upload Time Taken below:
HH : MM : SS
Get the Report
9:41
Lessons Report
You have completed 14 lessons today
60%
of Content Lvl2
Dyslexia 5 40%
Dysgraphia 17 56%
Dyscalculia 4 62%
Start a Lesson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Screening Tests
Lessons
Reports
Child's Profile Summary

COLOSSUS 2.0

TEAM MEMBERS DETAILS

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